Double descent in linear models

Theory for Assignment 4 (SMML 2025/26, Prof. Cesa-Bianchi)

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How to use this document. Reading time is about 40 minutes. Sections 2–6 are the minimum theory to be able to explain at the oral exam. Section 7 links the theory to the gradient descent extension, Section 8 describes what we expect to see in the plots, Section 9 the experimental design. At the end: likely oral questions with answers, and solutions to the check questions. Every claim states its source; when a concept is **not** in the uploaded course materials (Cesa-Bianchi's notes 1–15) it is flagged explicitly as an external source.

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1 Notation and correspondences

Concept	Course (slides 9)	Assignment / paper	Code
number of training examples	m	n	`m`
number of features (complexity)	d	d (assignment) / N (paper)	`d`
design matrix	$X \in \mathbb{R}^{m \times d}$		`X`
labels	$y \in \mathbb{R}^m$		`y`
ridge parameter	α	λ (other texts)	`alpha`
interpolation threshold	$d = m$	$N = n$	

Careful: in the paper d is the dimension of the *input* x, while the number of features/parameters is N and the number of samples is n (p. 2 and p. 4).

In the report we state once: "we follow the course notation; the number of examples n in the assignment corresponds to m". The rows of X are the examples: $X^\top = [x_1, \dots, x_m]$ (slides 9, p. 5). A linear predictor for regression is $h(x) = w^\top x$ with $w \in \mathbb{R}^d$ (slides 9, p. 1 and p. 4). The intercept is handled by adding a constant feature equal to 1 (slides 9, p. 1) or by centering the data.

2 Statistical setup: risk, training error, test error

Source: slides "4. statRisk" (p. 1–4) and "1. intro" (p. 3–5).

- Every example (x, y) is an independent draw from a fixed but unknown distribution $\mathcal{D}$ (statRisk, p. 1). Training and test sets are therefore statistical samples.
- **Statistical risk** of a predictor h with respect to the loss ℓ (statRisk, p. 1): $\ell_{\mathcal{D}}(h) = \mathbb{E}\big[\ell(h(X), Y)\big]$. For us $\ell(\hat{y}, y) = (\hat{y} - y)^2$ and $h(x) = w^\top x$.
- **Bayes optimal predictor** f^*: the best possible one given $\mathcal{D}$. For the square loss $f^*(x) = \mathbb{E}[Y \mid X = x]$ and the **Bayes risk** is the expected conditional variance of the label (statRisk, p. 1–2). In our synthetic dataset, with $y = w^{\star\top} x + \varepsilon$ and noise variance σ^2, the Bayes risk is exactly σ^2: **no model can have expected test error below** σ^2. We will use it as a reference line in the plots.
- **Training error**: $\ell_S(h) = \frac{1}{m} \sum_{t=1}^{m} \ell(h(x_t), y_t)$; for us $\ell_S(w) = \frac{1}{m} \|Xw - y\|^2$. ERM picks the predictor minimizing it (intro, p. 4).
- **Test error**: the same average on an independent test set S'; it estimates the risk $\ell_{\mathcal{D}}(h)$, and by Chernoff–Hoeffding the estimate is accurate when the test set is large (statRisk, p. 3–4). The test set is never used for training or for choosing hyperparameters.

Overfitting and underfitting in the course. Intro, p. 5: if the algorithm returns a predictor with high training error it is *underfitting*; with low training error but high test error it is *overfitting*. The course notes that overfitting is expected when the predictor class is too rich compared to m ("$\log_2 |F| \gg m$", intro p. 5). **Double descent is precisely a case where this classical intuition is not enough**: for $d \gg m$ the training error is 0 but the test error goes down again.

3 Least squares with $d \leq m$: the formula in the slides

Source: slides 9, p. 4–5.

The linear regression predictor is the ERM with respect to the square loss:

$$w_S = \operatorname*{argmin}_{w \in \mathbb{R}^d} \|Xw - y\|^2 .$$

$F(w) = \|Xw - y\|^2$ is convex, so the minimum is where the gradient vanishes:

$$\nabla F(w) = 2X^\top (Xw - y) = 0 \quad \iff \quad X^\top X\, w = X^\top y \qquad \text{(normal equations)}.$$

If $X^\top X$ is invertible:

$$\boxed{w_S = (X^\top X)^{-1} X^\top y}$$

When is it invertible? The slides say: if and only if $x_1, \ldots, x_m$ span $\mathbb{R}^d$. At least d vectors are needed to span $\mathbb{R}^d$, so we need $m \geq d$.

Geometric interpretation. The vector of training predictions is $Xw_S = X(X^\top X)^{-1}X^\top y = Py$, where P is the orthogonal projection onto the column space of X. The more features we add, the larger that space and the closer Py gets to y: the training error **cannot increase** as d grows.

The case $d = m$. If X is square and invertible, $w_S = X^{-1}y$ and $Xw_S = y$: training error exactly 0. The model already **interpolates** at $d = m$. Hence the name "interpolation threshold".

4 Least squares with $d > m$: the minimum-norm solution

External source (not in the course materials): Moore–Penrose pseudo-inverse, standard linear algebra textbooks (e.g. Strang, Introduction to Linear Algebra*, chapter on the pseudo-inverse and SVD); use of the minimum-norm solution: Belkin et al. 2019, Sec. 2 (p. 4: "when the minimizer is not unique ... we choose the minimizer whose coefficients have the minimum ℓ_2 norm") and p. 10 ("minimum norm linear regression in the feature space").*

4.1 Why the slides' formula no longer works

$X^\top X$ is a $d \times d$ matrix, but its rank equals the rank of X, which is at most $\min(m, d) = m$. If $d > m$ then $\operatorname{rank}(X^\top X) \leq m < d$: the matrix is **singular** and $(X^\top X)^{-1}$ does not exist. Equivalently: m vectors cannot span $\mathbb{R}^d$ when $m < d$.

4.2 Infinitely many interpolating solutions

With $d > m$ (and linearly independent rows of X, the typical case with continuous data), the system $Xw = y$ has m equations and d unknowns: it has **infinitely many** exact solutions. If w_0 is a solution, so is $w_0 + z$ for every z in the null space of X ($Xz = 0$), which has dimension $d - m$. All of them have zero training error.

4.3 Which one to pick: the minimum-norm solution

$$\boxed{w^+ = \operatorname{argmin}\{\|w\| : Xw = y\} = X^\top (XX^\top)^{-1} y}$$

Check that it interpolates: $Xw^+ = XX^\top (XX^\top)^{-1} y = y$. ($XX^\top$ is $m \times m$ and invertible when the rows of X are independent.)

Why it has minimum norm (short proof, know how to do it). $w^+ = X^\top a$ with $a = (XX^\top)^{-1} y$, so w^+ is a linear combination of the rows of X. Any other solution is $w = w^+ + z$ with $Xz = 0$, i.e. z is orthogonal to every row of X, hence to w^+. By Pythagoras:

$$\|w\|^2 = \left\|w^+\right\|^2 + \|z\|^2 \geq \left\|w^+\right\|^2 .$$

4.4 One formula for everything: the pseudo-inverse

The pseudo-inverse X^+ is computed from the SVD $X = U\Sigma V^\top$: $X^+ = V\Sigma^+ U^\top$, where Σ^+ inverts the non-zero singular values and leaves the others at zero. Then:

- if $d \leq m$ and X has full rank: $X^+ y = (X^\top X)^{-1} X^\top y = w_S$ (the slides' formula);
- if $d > m$ and X has rank m: $X^+ y = X^\top (XX^\top)^{-1} y = w^+$.

So in numpy `w = np.linalg.pinv(X) @ y` works for **every** d. It is linear algebra, not a library "fit", so it respects the "from scratch" rule (assignment FAQ, p. 16).

Numerical notes (important for the code).

- Do not use `np.linalg.inv(X.T @ X)`: forming $X^\top X$ squares the condition number, and near $d = m$ the results become numerically unreliable.
- `np.linalg.pinv` only zeroes singular values below a tiny threshold (`rcond`, about 10^{-15} relative to the largest). Near $d = m$ the small singular values are *not* zeroed: `pinv` does not regularize, so the peak stays visible. This is the behaviour we want to show.

4.5 Minimal numerical example

$m = 1$, $d = 2$: a single example $x = (1, 1)$, $y = 2$. The solutions of $w_1 + w_2 = 2$ are infinitely many: $(2, 0)$, $(0, 2)$, $(1, 1)$, $(3, -1)\ldots$ The minimum-norm solution is $w^+ = x^\top (xx^\top)^{-1} y = (1, 1) \cdot \frac{2}{2} = (1, 1)$, with norm $\sqrt{2}$, versus norm 2 for $(2, 0)$. It spreads the weight evenly instead of "betting everything" on one feature.

5 Why the test error blows up at $d \approx m$

Source: slides 9, p. 5: "when $X^\top X$ is nearly singular, w_S is highly sensitive to perturbations of the training set. This instability increases the estimation error (or variance)". This is the hook to the course.

5.1 Intuition with a 2×2 example

$m = d = 2$, rows $x_1 = (1, 0)$, $x_2 = (1, \varepsilon)$ with small ε (almost parallel rows, X nearly singular). Labels $y = (1,\ 1 + \delta)$, where δ is a small noise. Solving $Xw = y$: $w_1 = 1$ and $w_1 + \varepsilon w_2 = 1 + \delta$, so

$$w_2 = \delta / \varepsilon .$$

With $\varepsilon = 0.01$ and noise $\delta = 0.1$ we get $w_2 = 10$: a small noise produces a huge weight. On a new point, e.g. $x = (0, 1)$, the prediction is 10 instead of roughly 0. That is exactly the peak.

5.2 General version with eigenvalues

Let $\lambda_1 \geq \cdots \geq \lambda_d \geq 0$ be the eigenvalues of $X^\top X$ and $v_1, \ldots, v_d$ the (orthonormal) eigenvectors. The minimum-norm solution reads

$$w^+ = \sum_{i:\, \lambda_i > 0} \frac{v_i^\top X^\top y}{\lambda_i}\, v_i.$$

Assume noisy linear labels: $y = Xw^\star + \varepsilon$. The noise part is $\sum_i \frac{v_i^\top X^\top \varepsilon}{\lambda_i} v_i$; each term has norm $\frac{|u_i^\top \varepsilon|}{\sqrt{\lambda_i}}$, with $u_i = Xv_i/\sqrt{\lambda_i}$ a unit vector. **The noise is divided by $\sqrt{\lambda_i}$.**

- $d \ll m$: X is "tall and thin", all eigenvalues are large (about m for standardized features): the noise is damped.
- $d \approx m$: X is nearly square and random, the smallest eigenvalue is very close to 0: the noise is amplified, $\|w\|$ blows up, and so does the test error.
- $d \gg m$: X is "wide", the m non-zero eigenvalues move away from 0 again and the noise is spread over many directions: $\|w^+\|$ and the test error decrease. This is the **second descent**.

External source (not in the course materials): the behaviour of the smallest eigenvalue of a random matrix as a function of d/m is a random matrix theory result (Marchenko–Pastur law). We do not prove it: we show it empirically by plotting $\|w\|$ against d, which is easier to defend at the oral.

5.3 An exact formula for the classical regime (optional)

External source (not in the course materials): Hastie, Montanari, Rosset, Tibshirani, "Surprises in high-dimensional ridgeless least squares interpolation", Annals of Statistics 2022.

If $x \sim \mathcal{N}(0, I_d)$, $y = w^{\star\top} x + \varepsilon$ with noise variance σ^2 and $d < m - 1$, then

$$\mathbb{E}\big[\|w_S - w^\star\|^2\big] = \sigma^2 \frac{d}{m - d - 1},$$

which **diverges as** $d \to m - 1$. No need to prove it; it is useful to say at the oral that the peak is not a numerical artefact but a statistical phenomenon predicted by theory.

6 Ridge regression

Source: slides 9, p. 5.

6.1 Formula

$$w_{S,\alpha} = \operatorname*{argmin}_{w \in \mathbb{R}^d} \|Xw - y\|^2 + \alpha \|w\|^2, \qquad \alpha > 0.$$

Gradient: $2X^\top(Xw - y) + 2\alpha w = 0 \;\Rightarrow\; (X^\top X + \alpha I)\, w = X^\top y$, hence

$$\boxed{w_{S,\alpha} = (\alpha I + X^\top X)^{-1} X^\top y}$$

6.2 Why it removes the peak

The eigenvalues of $\alpha I + X^\top X$ are $\alpha + \lambda_i \geq \alpha > 0$ (slides 9, p. 5): the matrix is always invertible, for every d, including $d > m$. In the eigenvector basis:

$$w_{S,\alpha} = \sum_i \frac{v_i^\top X^\top y}{\lambda_i + \alpha} v_i.$$

We divide by $\lambda_i + \alpha$ instead of λ_i: even when $\lambda_i \approx 0$ the denominator stays at least α, so the noise cannot be amplified without bound. The peak is damped or disappears.

6.3 The price: bias

- $\alpha \to 0$: we recover the minimum-norm solution w^+ (also for $d > m$; along directions with $\lambda_i = 0$ the numerator $v_i^\top X^\top y$ is 0). *Standard fact, not in the slides.*
- $\alpha \to \infty$: $w_{S,\alpha} \to 0$, the predictor ignores the data (slides 9, p. 5: α controls the bias).
- In between: α trades variance (instability) for bias (underfitting). With too large an α the test error goes up *for every* d.

Source: Belkin et al. 2019, p. 10: regularization "can both prevent interpolation and change the effective capacity of the function class, thus attenuating or masking the interpolation peak".
External source (not in the course materials): Nakkiran, Venkat, Kakade, Ma, "Optimal regularization can mitigate double descent", ICLR 2021: with the optimal α the test error curve becomes monotone.

6.4 Computing it in practice

- Primal: `np.linalg.solve(X.T @ X + alpha * np.eye(d), X.T @ y)` ($d \times d$ system).
- Dual, cheaper for $d > m$: $w_{S,\alpha} = X^\top (XX^\top + \alpha I)^{-1} y$ ($m \times m$ system). The two forms agree because $X^\top (XX^\top + \alpha I) = (X^\top X + \alpha I) X^\top$.
- `solve` is preferable to computing the explicit inverse (more stable and faster).

6.5 Choosing α without data leakage

- For the main plot: a few curves with fixed α (e.g. $10^{-3}, 10^{-1}, 10$) next to the minimum-norm curve. No data-driven choice, no leakage.
- Time permitting: α chosen for each d on a validation split (or K-fold cross-validation) taken **only** from the training set (slides "6. crossVal", p. 2–3: tuning on a validation set; p. 3–4: nested CV), as in the professor's `CrossValidation.ipynb` notebook. The test set is never used for selection.

7 Link with gradient descent (extension)

Source: slides 10 (online gradient descent) for the algorithm; the minimum-norm result is external (see below).

Gradient descent on $F(w) = \|Xw - y\|^2$ started at $w_1 = 0$:

$$w_{k+1} = w_k - \eta\, 2X^\top (Xw_k - y).$$

- Every update is $X^\top(\dots)$, a linear combination of the rows of X. Starting from 0, w_k always stays in the row space.

- With a small enough step size ($\eta < 1/\lambda_1$), for $d > m$ GD converges to a solution that interpolates *and* lies in the row space: by the proof in Section 4.3, this is exactly w^+.
- Stopping before convergence (early stopping) has an effect similar to ridge: directions with small λ_i are learned last, so the noise along them is not amplified.

Source: implicit bias of GD towards minimum norm: Belkin et al. 2019, p. 11 ("SGD initialized at zero also converges to the minimum norm solution").

The extension is done **only** once the report and repo are complete.

8 What we expect in the plots

8.1 The classical U-curve and double descent

- **Classical regime** ($d < m$): the course decomposes the risk of h_S into *estimation error* (variance) + *approximation error* (bias) + Bayes risk (statRisk, p. 4). Adding features lowers the bias but raises the variance; overfitting when variance dominates, underfitting when bias dominates (statRisk, p. 6). Near $d = m$ the variance diverges: this is the U-curve.
- **Interpolation threshold** ($d = m$): training error reaches 0, test error peaks.
- **Over-parameterized regime** ($d > m$): training error is always 0, the test error decreases again and can go below the classical minimum (paper, Fig. 1, p. 3). The paper's explanation (p. 3 and p. 6): beyond the threshold, richer classes contain interpolants with smaller norm, i.e. "simpler" ones (Occam's razor); the norm peaks at the threshold and then decreases (Fig. 2, p. 5). Classical theory, which predicts only the U, does not explain this part: this is the point of the paper.

8.2 Planned plots

1. Training and test error against d (minimum norm), vertical line at $d = m$, logarithmic y axis (the peak spans orders of magnitude).
2. On the synthetic data, a horizontal line at the Bayes risk σ^2 (Section 2) as a reference.
3. $\|w\|$ against d: it must peak at the same place (evidence for the mechanism of Section 5).
4. Test error of minimum norm and ridge (several α) on the same plot.
5. The same plots for the real dataset, and a comparison between the two datasets.

9 Experimental design (preview)

Relation to the paper. The paper shows the phenomenon with random Fourier and random ReLU features (Sec. 2 and App. C.1–C.2), neural networks (Sec. 2) and forests/boosting (Sec. 3), on MNIST and other classification datasets. It has no "pure" linear regression experiment, but the random features are trained by minimum-norm linear regression in feature space (p. 10): our experiment is its simplest version.

9.1 Synthetic dataset: noisy linear labels

External source (not in the course materials): "weak features" model: Belkin, Hsu, Xu, "Two models of double descent for weak features", 2019.

- $x \sim \mathcal{N}(0, I_D)$ with large D (e.g. $D = 1000$), $y = w^{\star\top} x + \varepsilon$, Gaussian noise.

- The model with complexity d uses only the first d features. For $d < D$ the unused features act as extra noise (bias); for large d the model has access to all the signal.
- Fixed m (e.g. $m = 100$), d from 1 to D, denser grid around $d = m$.
- Mean (or median) over several repetitions with different seeds, with a variability band.

9.2 Real dataset with random features

- A regression dataset with few features (e.g. California Housing, 8 features, also in the professor's repo).
- To push d past m: random features $\phi(x) = \max(0, Wx)$ (ReLU) with a fixed Gaussian W, or random Fourier features as in the paper. The model stays **linear in the parameters** w. The course calls this idea *feature expansion*: adding new features built from the original ones to reduce the bias of linear predictors (slides "11. kernels", p. 1).
- We subsample m training examples (e.g. $m = 200$) and keep a separate test set.

9.3 Rules against data leakage

- Train/test split *before* any pre-processing.
- Mean and standard deviation for standardization computed **only** on the training set and applied to the test set.
- The random matrices W do not depend on the data, so they create no leakage, but they must be generated with a fixed seed.
- Hyperparameters (e.g. α) chosen only with training data.

10 Likely oral questions, with short answers

1. **What is double descent?** As a function of complexity, the test error decreases, rises to a peak at the interpolation threshold ($d = m$), then decreases again in the over-parameterized regime.
2. **Why is the peak exactly at $d = m$?** There X is nearly square, $X^\top X$ has an almost-zero eigenvalue, the solution divides the noise by that eigenvalue and $\|w\|$ blows up (slides 9, p. 5 and Section 5).
3. **Why is the training error 0 for $d \geq m$?** With $d \geq m$ and independent rows, $Xw = y$ has at least one exact solution: the model interpolates.
4. **Which solution do you use for $d > m$ and why?** The minimum-norm X^+y: it is unique, interpolates, is the limit of ridge as $\alpha \to 0$ and the point GD converges to when started at 0.
5. **How did you compute X^+?** With `np.linalg.pinv`, based on the SVD. It is linear algebra, not a fit.
6. **What does ridge do to the peak?** It replaces λ_i with $\lambda_i + \alpha$ in the denominator: no noise amplification, peak damped or removed, at the cost of some bias.
7. **How did you choose α?** Fixed values compared side by side, and/or validation on the training set only. Never on the test set.

8. **How do you vary d on the real dataset?** With random features; the model stays linear in the parameters.

9. **How do you avoid data leakage?** Split first, standardization statistics from the training set only, hyperparameters from the training set only.

10. **Why does double descent contradict classical theory?** Classical theory predicts that beyond some point more complexity always hurts generalization (U-curve). In the $d > m$ regime the opposite happens.

11. **Weak points of your work?** Linear models only; the real dataset needs random features, so the result depends on how they are built; few repetitions because of time constraints; arbitrary choice of m.

11 Solutions to the check questions

Try to answer on your own first, then compare.

Question 1. Why can't the slides' formula $(X^\top X)^{-1}X^\top y$ be used for $d > m$, and what does `pinv(X) @ y` compute?

Answer. $X^\top X$ is $d \times d$ but has rank at most $m < d$, so it is singular and not invertible (equivalently, m vectors cannot span $\mathbb{R}^d$, slides 9 p. 4). The system $Xw = y$ has infinitely many interpolating solutions. `pinv(X) @ y` returns the one with minimum Euclidean norm, $X^\top(XX^\top)^{-1}y$: it is unique, lies in the row space of X, and every other solution has a larger norm (Pythagoras). For $d \leq m$ it coincides with the slides' formula.

Question 2. Why does ridge have no peak at $d = m$?

Answer. In the minimum-norm solution the components of y are divided by the eigenvalues λ_i of $X^\top X$; at $d \approx m$ the smallest one is almost 0 and the noise blows up. Ridge inverts $\alpha I + X^\top X$, whose eigenvalues are $\lambda_i + \alpha \geq \alpha > 0$ (slides 9 p. 5): the denominator never drops below α, so $\|w\|$ and the test error stay bounded. The price is bias: with too large an α, $w \to 0$ and the model underfits.

12 Sources

- N. Cesa-Bianchi, notes "Introduction" (1. intro), p. 3–5; "Statistical learning" (4. statRisk), p. 1–6; "Hyperparameter tuning and risk estimates" (6. crossVal), p. 2–4; "Kernels" (11. kernels), p. 1.
- N. Cesa-Bianchi, slides "Linear predictors" (9. linear), version of 10/02/2026, p. 4–5.
- N. Cesa-Bianchi, slides "Online Gradient Descent" (10. online), version of 24/02/2026.
- M. Belkin, D. Hsu, S. Ma, S. Mandal, "Reconciling modern machine learning practice and the classical bias-variance trade-off", PNAS 2019 (arXiv:1812.11118). arXiv v2 (10/09/2019); page numbers refer to that PDF.
- M. Belkin, D. Hsu, J. Xu, "Two models of double descent for weak features", 2019 (arXiv:1903.07571). *External.*
- T. Hastie, A. Montanari, S. Rosset, R. Tibshirani, "Surprises in high-dimensional ridgeless least squares interpolation", Annals of Statistics 2022. *External.*

- P. Nakkiran, P. Venkat, S. Kakade, T. Ma, "Optimal regularization can mitigate double descent", ICLR 2021. *External.*
- Assignment "ML projects 25/26", Assignment 4 (p. 8–9) and FAQ (p. 16).